CRAM

User Manual

A guide to operating the Crisis Role Availability Manager during routine use and live incidents.

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1. Getting started

> *CRAM is a single-file web application for managing crisis committee roles and substitution chains. Everything runs inside the browser — no server, no cloud, no installation required.*

1.1 Opening the tool for the first time

Open `crisis-role-manager.html` with any modern browser. A double-click on the file in your operating system is usually enough; macOS, Windows, and Linux all hand HTML files to the default browser.

On first launch, CRAM loads a small sample configuration — a basic crisis committee for cybersecurity incidents with around a dozen roles. This is there so you can immediately explore the interface; it is not meant for real use.

Your committee administrator will normally provide a proper configuration, either as a JSON file to import or by transferring it via QR code from their own instance. See **Chapter 4** for the three available transfer channels.

1.2 Installing as a progressive web app

CRAM can be installed as an app on any modern platform. The installed version runs in its own window without the browser's URL bar and tabs, and it appears as a regular application in your launcher, dock, or home screen.

Platform	How to install
Chrome / Edge (desktop)	Click the install icon on the right side of the address bar
Chrome on Android	Menu → "Add to Home screen"
Safari on iOS / iPadOS	Share menu → "Add to Home Screen"
Firefox (desktop)	Limited support — installation works on some builds; use the browser tab otherwise

> **STATE PERSISTENCE**
>
> The installed app and the browser tab share the same storage area (localStorage) as long as they open the same file or URL. Switching between them does not lose data.

1.3 The interface at a glance

The main screen has three regions: a **header bar**, the **organisation chart** in the centre, and a **sidebar** on the right with three tabs.

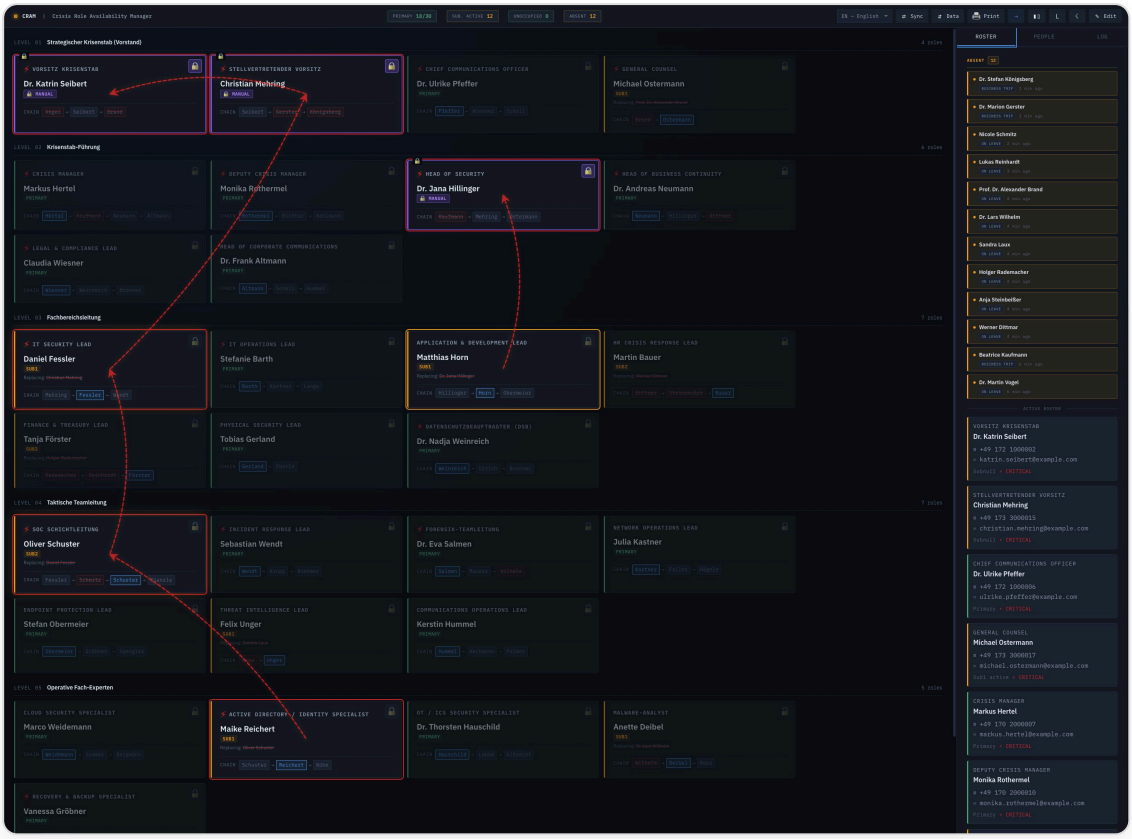

Figure 1 — Main view with the enterprise demo configuration. Six levels, thirty roles, cascade arrows tracing live substitutions. The right sidebar shows the current absence list.

The header contains, from left to right:

- **Brand and version indicator**, so you always know which release you are running
- **Four status pills**: Primary (how many primaries are currently present), Sub. active (how many substitutes have stepped in), Unoccupied (roles with no one available — this pill turns red if a critical role is affected), and Absent (total number of unavailable people)
- **Language selector**
- **Action buttons**: Sync (⇄), Data (⇵), Print (🖶), Edit mode toggle (✎), and theme switcher (☀/☾)

2. Reading the organisation chart

> *The centre of the screen shows the entire committee as a hierarchical chart. Each role is displayed as a card; cards are grouped into levels.*

2.1 Role cards

Every role card contains, from top to bottom:

- A small ⚡ symbol if the role is marked as **critical**. Critical roles take priority in the statistics and are highlighted in printouts.
- The **role name** in uppercase letters — this is the function, not a person's title.
- The **current occupant**'s name, marked with a rank badge:
  - `PRIMARY` — the planned person is present and holds the role
  - `SUB1`, `SUB2`, … — a substitute has taken over because the planned person is unavailable
- A **Replacing:** line, visible only when a substitute has stepped in. The person being replaced is shown with strikethrough.
- The **CHAIN** row at the bottom showing the full substitution sequence. Unavailable people appear with strikethrough.

A lock icon (🔒) in the top-right corner of a card means the role currently has a *manual assignment* — someone has pinned a specific person to that role, overriding the automatic logic. See Section 3.3.

2.2 Substitution chain

Each role has an ordered list of people who can fill it. The first person is the **primary**; subsequent people are substitutes in decreasing order of preference. If the primary is unavailable, CRAM walks down the list and picks the first available person.

The chain is visible at the bottom of every role card. Names are shown in small pills, connected by arrows. The currently active person is highlighted; unavailable people are struck through.

2.3 Cascade arrows

When one person covers for another on a *different level*, CRAM draws an animated dashed arrow between the two affected role cards. This makes it immediately visible which substitution chains are currently under stress — useful during a briefing when the committee lead needs to know at a glance which positions are being exercised.

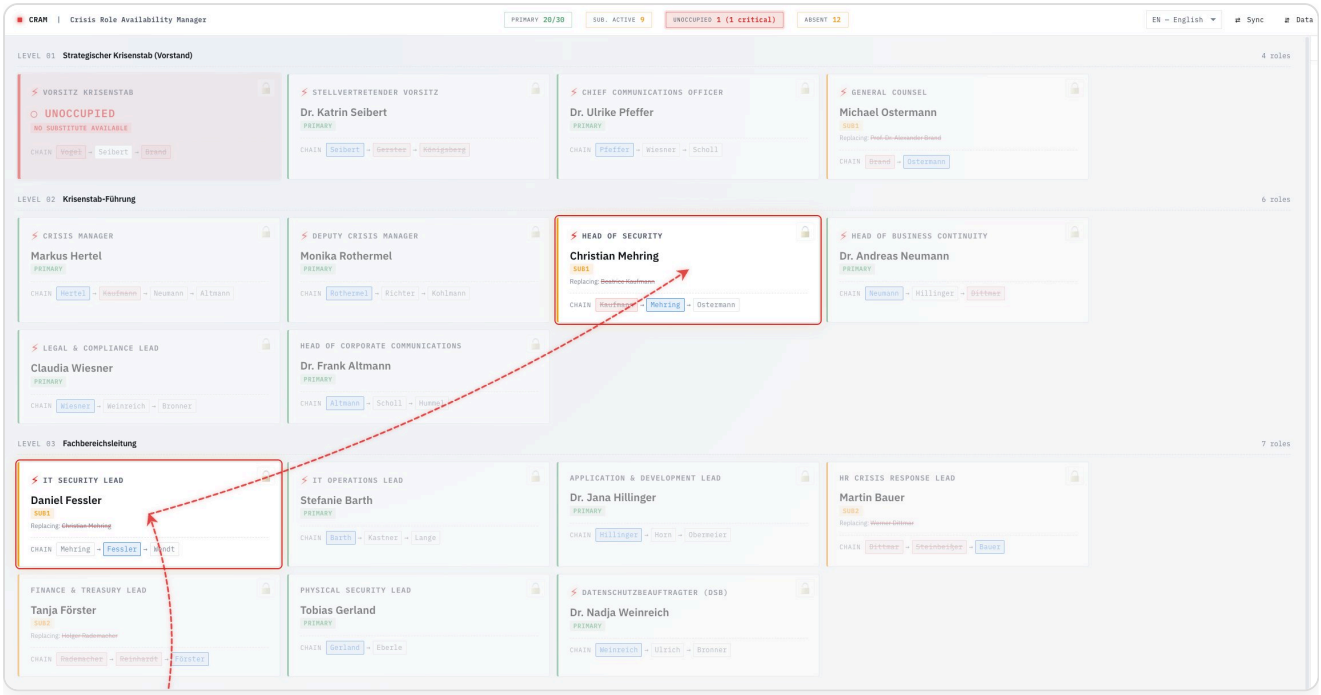

Figure 2 — Light theme. A critical role is unoccupied (red panel, top-left). The arrows trace how substitutes on other levels are now covering — a visible indicator that the chains are being stressed.

2.4 Visual legend

CRAM uses colour and animation throughout the interface to signal the current state of every role and cascade. The following tables cover every visual cue the tool emits.

Role card left border — current state of the role

Colour	State	Meaning
■ Green	Primary	Planned primary occupant is active
■ Yellow	Substitute (Sub1)	First substitute has taken over
■ Dark orange	Substitute deeper (Sub2+)	Second-tier or later substitute is active
■ Red	Unoccupied	Nobody in the chain is available
■ Red + pulsing	Unoccupied + critical	Critical role is unoccupied — attention required

Role card border and surroundings — special conditions

Signal	Meaning
Purple solid border + 🔒 badge	Role has an active manual assignment
Purple dashed border + struck-through name	Manually assigned person has become unavailable
Yellow outline around card	Card is part of a non-critical cascade
Yellow outline + glow	Cascade target (non-critical) — substitute is actively standing in here
Red outline around card	Card is part of a cascade into a critical role
Red outline + glow	Cascade target on a critical role
Dimmed to 50% opacity	Cascade view active; this card is not part of any cascade

Cascade arrows (only visible when cascade view is enabled)

Colour	Animation	Meaning
■ Yellow dashed	Slow flow	Substitution into a non-critical role
■ Red dashed, thicker	Faster pulse with red glow	Substitution into a **critical** role — watch closely

> **WHICH COLOUR WINS?**
>
> The arrow colour is determined by the **target role** (where the person is standing in), not by the person's home role. A green substitute stepping into a red-critical role produces a red arrow; a primary from a critical role covering a non-critical role produces a yellow arrow.

Header status pills

Colour	Meaning
Grey (neutral)	Informational count, no action required
■ Yellow outline	Active but non-critical — substitutes have stepped in
■ Red + pulsing	At least one critical role is unoccupied

Sidebar accents

Element	Colour	Meaning
Roster entry — left border	■ Green	Role held by primary
Roster entry — left border	■ Yellow	Role held by substitute
Roster entry — left border	■ Dark orange	Role held by deeper substitute
Roster entry — left border	■ Red	Role unoccupied
People list entry — left border	■ Yellow	Person is currently unavailable

2.5 Status pills in the header

The four status pills give an immediate overview of the current state:

Pill	What it counts	Colour logic
Primary	Roles held by the planned primary, out of the total number of roles	Neutral grey
Sub. active	Roles where a substitute has stepped in	Orange outline when greater than zero
Unoccupied	Roles with no available person in the chain	Red when a critical role is affected; orange otherwise
Absent	Total number of people currently unavailable	Neutral grey

3. Day-to-day operation

> *Outside edit mode — the normal state — clicking on anyone in the chart opens the availability dialog. This is the single most common action during a live incident.*

3.1 Marking a person unavailable

Click the person's name on any role card. The "Mark as unavailable" dialog opens. Pick a reason from the drop-down:

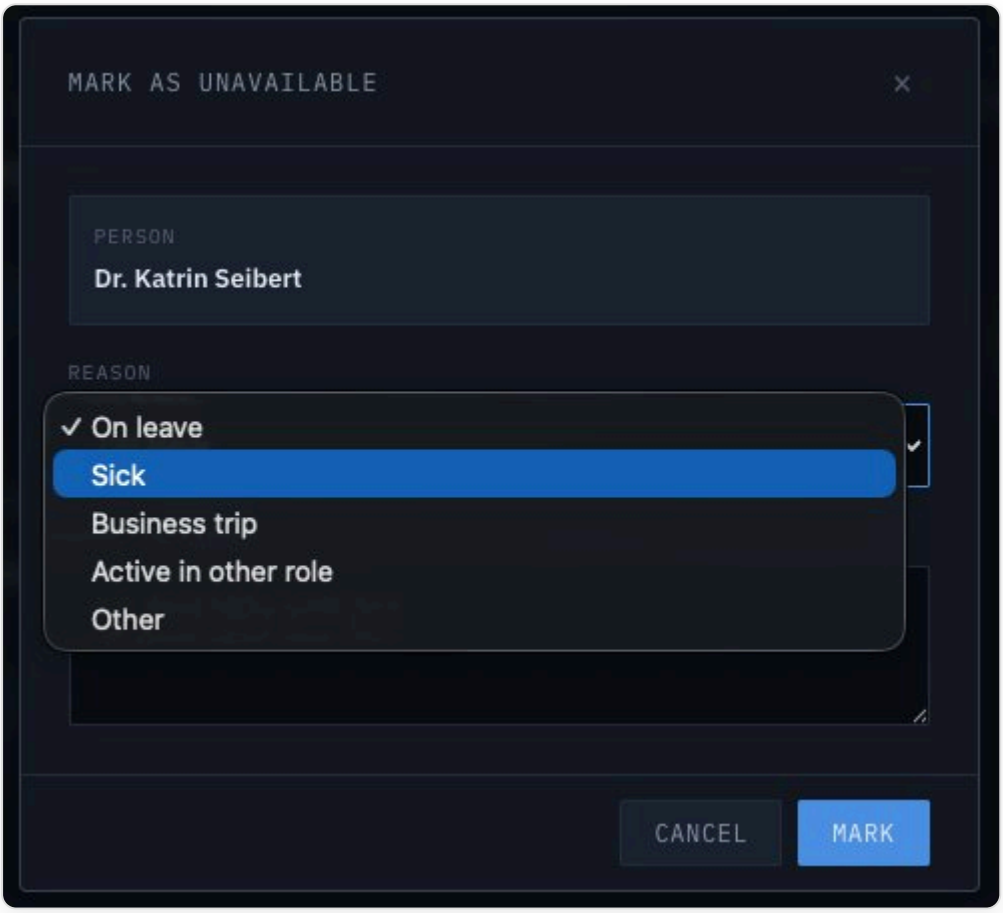

Figure 3 — Marking someone as unavailable. A reason category is mandatory; the note field is optional but recommended during active incidents.

The reason categories are:

- **On leave** — scheduled absence, holiday
- **Sick** — short-notice medical absence
- **Business trip** — travelling, potentially reachable but not available for the committee
- **Active in other role** — the person is busy with another task and cannot be pulled into this committee right now
- **Other** — anything that doesn't fit the above

Add a note if it helps the next person reading the log — for instance an expected return time, a handover instruction, or the reason for "Other".

After confirmation, the person is marked absent. Every role they were filling is automatically re-resolved. If the person was primary somewhere, the first available substitute takes over; cascade arrows update accordingly.

3.2 Releasing an absence

To bring someone back from the absence list, click their name in the sidebar (Roster or People tab) and confirm in the dialog. The person becomes available again; roles in which they are primary revert to them automatically, unless a manual assignment has been placed in the meantime.

3.3 Manual assignment

Sometimes the automatic substitution logic isn't what you want during an actual incident. Examples:

- The planned primary is technically available but is already tied up with another urgent matter
- The committee has decided operationally that a specific substitute should take a role, bypassing the planned succession
- A senior person wants to delegate a role they would normally hold

In these cases, use **manual assignment**. In edit mode (or via the pin icon on a role card), you can pick any person from the entire directory and pin them to a role.

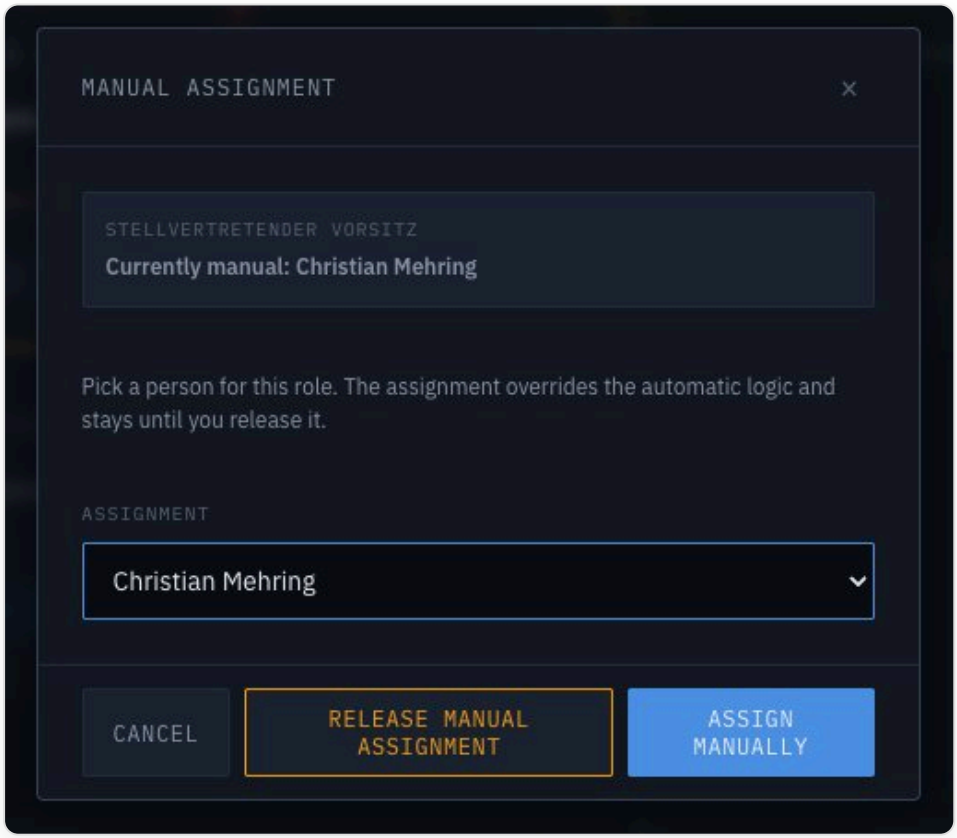

Figure 4 — The manual assignment dialog. The assignment overrides automatic logic and stays until explicitly released.

The manually assigned role is marked with a 🔒 badge. The assignment stays in place until it is explicitly released via "Release manual assignment" in the same dialog. Manual assignments are included in sync transfers, so the state propagates to other devices.

3.4 Using the sidebar

The right sidebar has three tabs:

Roster

The default view. Shows all currently absent people at the top, then the active occupants grouped by their role. A handy reference when you need to look up a phone number quickly.

People

All persons in the directory, listed alphabetically with their current availability and role count. Useful for marking absences without having to find the right role card first.

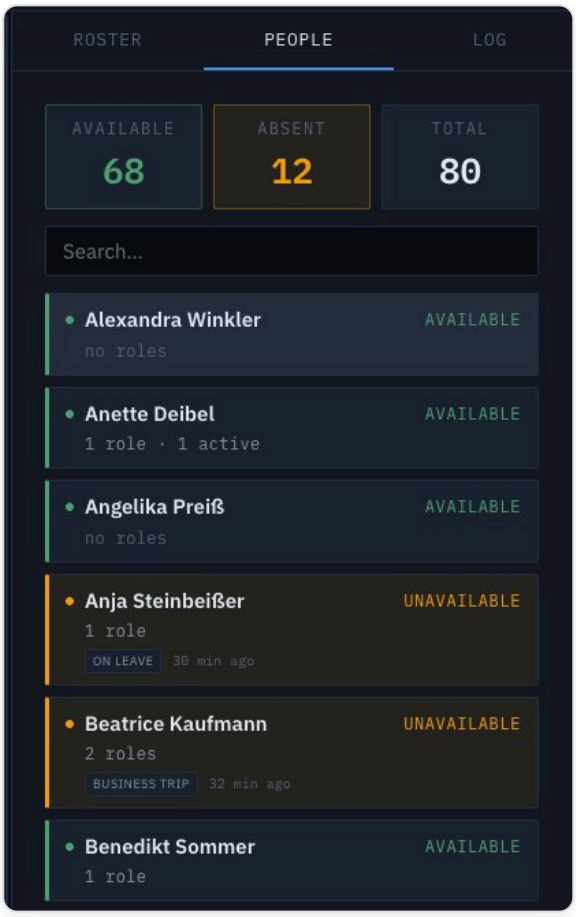

Figure 5 — People tab with availability stats and search filter. Available and absent summaries at the top.

Log

The 30-day audit log — every absence, release, manual assignment, configuration edit, and sync transfer is recorded here with a timestamp. Entries older than 30 days are removed automatically.

Figure 6 — The audit log. Useful during post-incident reviews and when the committee needs to understand what happened and in what order.

4. Transferring data between devices

> *CRAM has three independent transfer channels. Each solves a different real-world problem; you will likely use all of them at different moments during committee work.*

4.1 Which channel to use

Channel	What it transfers	Requires	Typical situation
Sync code	Status only (absences, manual assignments)	Identical base configuration on both sides	Quick updates by phone, chat, or radio
QR transfer	Configuration, status, or both	Camera on receiving device; no network required	Initial distribution or when configurations differ
JSON export / import	Configuration, status, or both	A way to move a file (email, share, USB)	Archival, versioning, email distribution

> **IMPORTANT DISTINCTION**
>
> CRAM separates the **configuration** (the static structure of roles and people) from the **status** (who is currently unavailable and which manual assignments are active). Sync codes move only status. QR transfer and JSON can move either or both.

4.2 Sync code (phone)

A sync code is a short alphanumeric string that encodes the current status. It is formatted in groups of four characters so it is easy to read aloud. Typical length is 20 to 40 characters depending on the number of people in the directory and how many are absent.

The first four characters after the leading version digit form a **configuration fingerprint**. Both sides must have the same fingerprint for the code to apply; mismatched fingerprints are rejected before any change is made.

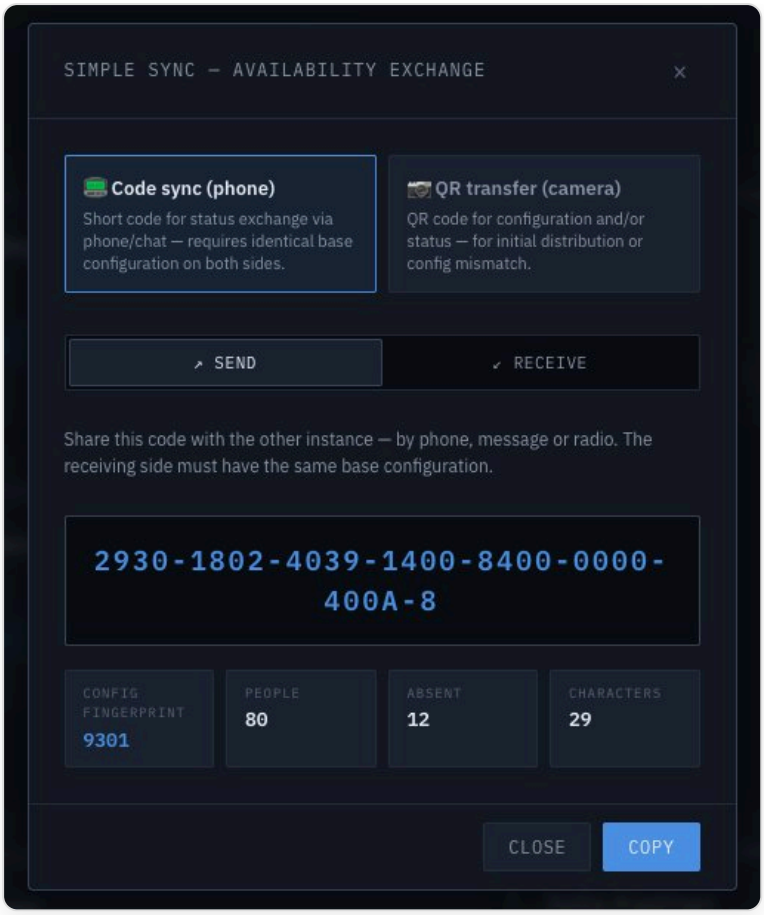

Figure 7 — Code sync view. The fingerprint and the character count are shown for verification.

Sending a sync code

1. Open the sync modal (⇄ button in the header)
2. Select the channel "📟 Code sync (phone)" if it isn't already selected
3. Make sure the "↗ Send" tab is active — the current code is shown immediately
4. Read the code aloud, or copy it to paste into a chat or radio log

Receiving a sync code

1. Open the sync modal, same channel as the sender
2. Switch to the "↙ Receive" tab
3. Enter the code — spaces, dashes, and capitalisation do not matter
4. The fingerprint is validated immediately and the entered fingerprint is displayed for visual comparison
5. Click "Apply" to commit

> **IF THE RECEIVING SIDE HAS NEVER HAD THIS CONFIGURATION**
>
> The sync code will be rejected. Use QR transfer or JSON import first to establish the same base configuration.

4.3 QR transfer (camera)

For device-to-device transfer of anything — configuration, status, or both together. The sender generates one or more QR codes; the receiver scans them with the camera.

Figure 8 — QR transfer. For small payloads (like status-only) a single QR suffices. Larger configurations produce a numbered sequence.

Sending

1. Sync modal → "📷 QR transfer (camera)" → "↗ Send" tab
2. Pick scope:
   - **Configuration only** — roles, levels, people (no status)
   - **Status only** — absences and manual assignments
   - **Configuration + status** — full snapshot
3. Click "Generate QR code(s)"
4. If the transfer fits into a single QR, it appears immediately
5. For multi-fragment transfers, use the navigation buttons or activate "▶ Auto-advance" — the QRs rotate every 2.5 seconds

Receiving

1. Sync modal → "📷 QR transfer (camera)" → "↙ Receive" tab

2. Click "Start camera" and grant camera permission
3. Point the camera at the sender's screen. The reticle shows the focus area; fragments are detected automatically
4. Progress dots at the bottom turn green as fragments are collected
5. Once every fragment is in, the preview appears with statistics about the incoming payload
6. Click "Apply" to commit; a log entry is written

> **BROWSER REQUIREMENTS**
>
> QR scanning needs the native `BarcodeDetector` API. Works on Chrome, Edge, and Safari. **Firefox does not have it** — Firefox users should use JSON import instead for initial setup. Camera access on mobile browsers requires HTTPS or `file://`; plain HTTP will not work.

4.4 JSON export and import

For archival, email distribution, and any case where a plain file is easier to handle than a camera-to-screen transfer.

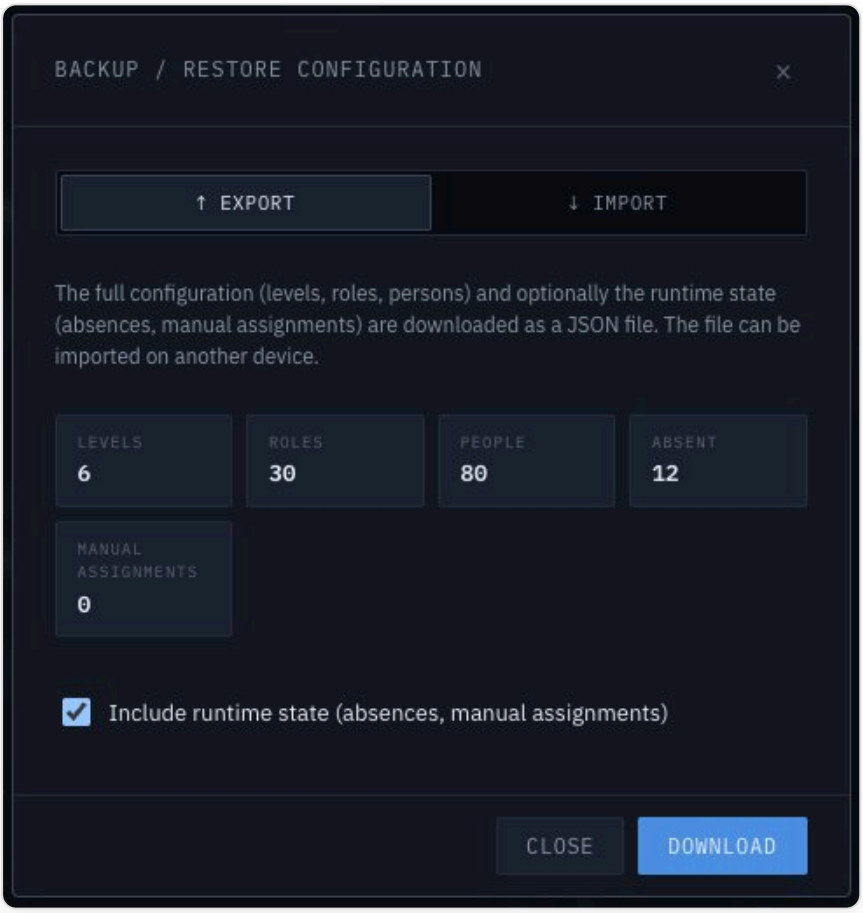

Figure 9 — Export dialog with summary statistics. The "Include runtime state" checkbox decides whether absences and manual assignments are part of the export.

Export

1. Data modal (⇵) → "↑ Export" tab
2. Tick "Include runtime state" if you want absences and manual assignments in the file

3. Click "Download" — the file is saved as `cram-export-YYYY-MM-DD-HH-MM-SS.json`

Import

1. Data modal → "↓ Import" tab
2. Select the JSON file
3. Review the preview (number of levels, roles, people)
4. Confirm — existing data is overwritten

> **IMPORT OVERWRITES**
>
> There is no merge; importing replaces the current configuration entirely. If you want to keep the existing state, export it first as a backup.

5. Printing and PDF output

> *CRAM can produce three different paper outputs, each suited to a different situation. All three work with A4, A3, or Letter, in portrait or landscape.*

5.1 Three templates

Template	Purpose	Best paper
Overview	Single-page wall chart. Roles grouped by level, each with its current primary and phone number in large type.	A3 landscape for committees of 20+ roles; A4 landscape for smaller
Role detail	Multi-page structured listing. Each role with full chain, phone numbers, and any manual assignment.	A4 portrait
People list	Alphabetical phone directory. Absent people called out in a separate section.	A4 portrait

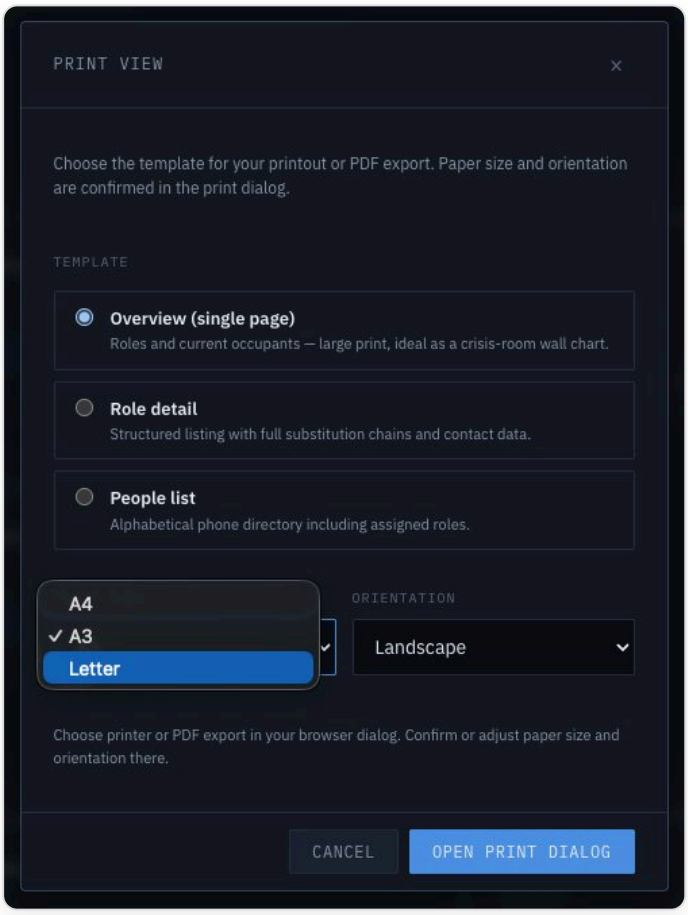

Figure 10 — Print dialog. Pick the template and paper size before handing off to the browser's print dialog, where you can choose "Save as PDF" if you don't want a physical print.

5.2 Paper size and orientation

The layout scales to fill the chosen page. Critical roles are always marked in red; regardless of theme on screen, printouts use black text on white background for readability and toner economy.

5.3 Using the output in a crisis room

Pragmatic tips from real use:

- **Overview template, A3 landscape, laminated**: Pin it to the wall of the crisis room and annotate with a dry-erase marker. Survives liquid accidents; cheap to replace.
- **Role detail, A4 portrait, in a folder**: One sheet per committee member. Folded in half fits in a shirt pocket.
- **People list**: A laminated A5 card at each desk in the crisis room with the phone directory.
- **Refresh before every scheduled test**: Print current versions so outdated phone numbers don't slip through.

6. Settings and personalisation

6.1 Organisation and print titles

Settings (⚙, available from the header while in edit mode) exposes two fields that affect labels and printouts:

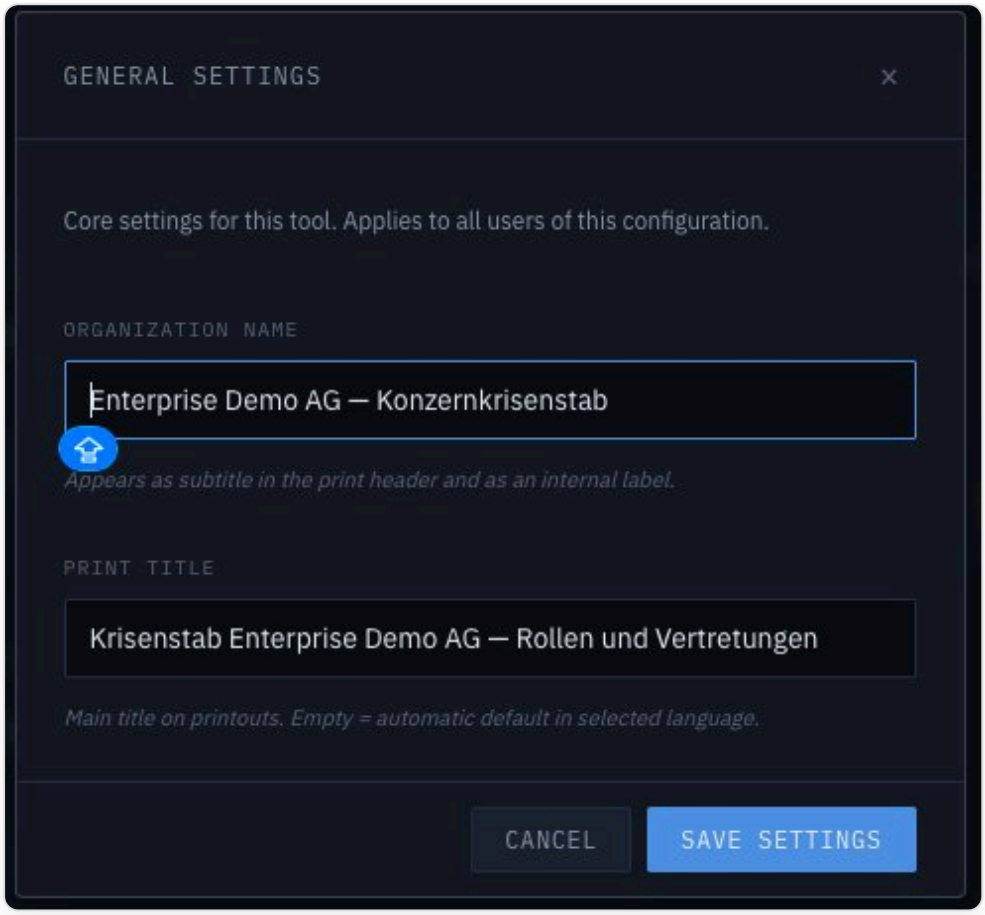

Figure 11 — The General Settings dialog with organisation name and print title.

- **Organisation name** — appears as a subtitle in the print header and as an internal label
- **Print title** — the main title on every printed page. If empty, a language-dependent default is used.

6.2 Languages

Five languages are shipped: German (Deutsch), English, Spanish (Español), French (Français), and Chinese (中文). The language switcher is in the header; the choice persists between sessions.

Figure 12 — Language switcher in the header.

6.3 Theme and display density

The ☀/☾ toggle switches between light and dark theme. The four size steps (S, M, L, XL) change the density of the organisation chart — useful on different monitors or when the chart gets large.

7. Audit log

> *The Log tab in the sidebar records every state-changing action over the past 30 days. Entries older than 30 days are removed automatically.*

Recorded events include:

- Availability changes (marked unavailable / released)
- Manual assignments (pinned and released)
- Configuration edits (role, level, or person modified)
- Sync events (incoming code, outgoing code, QR send, QR receive)
- Imports and exports

The log is stored locally on the device. There is no cross-device synchronisation of the log — each committee member sees their own record of events as they experienced them. After a live incident, the maintained instance of the committee administrator is usually the reference log.

> **POST-INCIDENT REVIEW**
>
> Exporting the log for a post-incident review isn't a built-in feature in 1.0-rc1; if you need it, taking screenshots of the Log tab while still on the committee administrator's instance is the pragmatic approach for now.

8. Troubleshooting

"The QR scanner doesn't activate"

On Firefox, the scanner is not available because the browser does not have the `BarcodeDetector` API. Use Chrome, Edge, or Safari — or use JSON import as a fallback.

If the scanner tries to start but the camera stays black, you are likely serving the file over plain HTTP. Camera access requires HTTPS, `localhost`, or a `file://` URL.

"My sync code is rejected"

The most common cause is a configuration fingerprint mismatch — the sender and receiver have different base configurations. Check the fingerprint displayed on both sides; they must match. If they don't, someone needs to send their configuration via QR or JSON first, then try the sync code again.

"Everything is gone after I reopen the file"

Your browser's localStorage is scoped to the file path (the origin). If the HTML file was moved or copied, opening the new location shows a fresh installation. Keep the file in a stable location, or host it on a server.

In private / incognito browsing mode, all data is discarded when the last tab closes. This is browser behaviour, not a bug.

"The Overview print breaks mid-page"

For committees with 30+ roles, A4 landscape is tight. Switch to A3 landscape for a clean single page, or use the Role Detail template which breaks naturally across pages.

"Two committee members have different data"

Expected. CRAM does not auto-sync. Before any meeting or test, use the sync code to align status (quick) or QR transfer to align the full state (thorough).

9. Reference card

> *The one-page reference for anyone who already knows the tool. Print this alone if you like.*

Header actions

⇄ Sync	Exchange state with another instance (code or QR)
⇵ Data	Import / export JSON
🖨 Print	Open the print template picker
✎ Edit	Toggle edit mode (add / modify roles, levels, people)
☀/☾	Toggle light / dark theme
S M L XL	Change density of the organisation chart

Common actions

Mark absent	Click any person's name in the chart
Release absence	Click the person in the absence list (Roster or People tab)
Manual assignment	Click the pin icon on a role card (edit mode)
Release manual assignment	Open the same dialog, click "Release manual assignment"
Quick sync by phone	⇄ → Send tab → read code aloud. Receiver: Receive tab → enter code
Config transfer	⇄ → QR transfer → Send → scope "Configuration + status"
Archive current state	⇵ → Export → tick "Include runtime state" → Download

Sidebar tabs

Roster	Active occupants grouped by role, current absences at top
People	Alphabetical directory with filter and availability summary
Log	30-day audit trail

Status pill colour logic

Grey	Informational count, no action needed
Orange outline	Active but non-critical — substitutes have stepped in
Red	At least one critical role is unoccupied — attention required

Browser compatibility

Feature	Chrome/Edge	Safari	Firefox
Core functionality	✓	✓	✓
QR code generation	✓	✓	✓
QR code scanning	✓	✓	—
PWA installation	✓	✓	partial
Camera over `file://`	✓	✓	✓

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`github.com/Artaiios/CRAM`